

ZEYA, Russian Federation

WMO: 313000

Lat: 53.7421N

Lon: 127.2760E

Elev: 755

StdP: 14.30

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.2	-28.5	-36.0	0.7	-31.5	-32.5	0.9	-27.4	15.1	11.4	13.3	8.4	0.0	220	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	86.4	68.3	83.2	66.8	80.2	65.9	71.5	81.3	69.7	79.1	68.0	76.9	3.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.4	107.2	75.9	66.6	100.6	74.2	64.8	94.6	72.8	35.9	81.3	34.4	79.0	32.9	76.8	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.1	11.0	9.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-34.2	91.9	4.1	3.5	-37.1	94.4	-39.5	96.4	-41.8	98.4	-44.7	100.9
			-34.3	74.3	4.0	1.6	-37.2	75.4	-39.6	76.4	-41.8	77.3	-44.7	78.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	30.9	-10.3	-1.3	16.4	36.6	51.6	64.2	69.3	64.1	50.6	32.0	5.8	-9.6	(6)
(7)		DBStd	29.80	9.95	9.13	10.46	7.79	7.07	6.33	5.08	5.68	7.57	9.67	11.82	10.40	(7)
(8)		HDD50	8642	1869	1437	1041	408	65	1	0	1	85	562	1325	1848	(8)
(9)		HDD65	12705	2334	1857	1506	852	418	87	19	85	435	1024	1775	2313	(9)
(10)		CDD50	1688	0	0	0	6	115	428	597	436	102	4	0	0	(10)
(11)		CDD65	276	0	0	0	0	3	64	151	56	2	0	0	0	(11)
(12)		CDH74	2642	0	0	0	4	97	792	1264	469	16	0	0	0	(12)
(13)		CDH80	711	0	0	0	0	17	225	385	84	0	0	0	0	(13)
(14)	Wind	WSAvg	3.0	2.0	2.6	3.8	4.6	4.3	3.1	2.7	2.5	2.7	3.2	2.4	2.2	(14)
(15)	Precipitation	PrecAvg	20.7	0.2	0.2	0.3	1.1	1.4	3.3	5.6	4.2	2.9	0.9	0.5	0.3	(15)
(16)		PrecMax	29.3	0.6	0.7	1.1	2.3	3.5	8.1	12.9	8.5	7.5	5.6	1.0	1.3	(16)
(17)		PrecMin	14.2	0.0	0.0	0.0	0.2	0.2	0.5	1.5	1.0	0.6	0.0	0.1	0.0	(17)
(18)		PrecStd	4.0	0.2	0.2	0.3	0.6	0.7	1.9	2.8	2.0	1.7	1.1	0.3	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.4	26.1	46.5	70.1	82.4	90.1	92.6	86.2	76.4	62.0	34.1	25.2	(19)
(20)		MCWB	17.1	21.4	36.1	50.4	59.8	67.5	69.6	70.0	62.8	49.8	29.9	20.7		(20)
(21)		2%	DB	14.1	21.7	40.9	61.7	76.3	85.7	87.4	82.2	71.5	55.5	30.7	19.7	(21)
(22)		MCWB	11.2	17.9	32.0	45.9	56.2	66.1	69.7	67.9	59.6	45.1	27.5	16.6		(22)
(23)		5%	DB	9.5	17.3	36.4	55.4	71.4	82.3	84.2	79.2	67.7	51.0	27.6	10.7	(23)
(24)		MCWB	7.3	14.2	29.2	42.4	53.6	64.2	68.7	66.6	57.1	42.0	24.4	8.6		(24)
(25)		10%	DB	5.3	13.2	32.7	50.9	66.7	78.7	80.9	76.1	64.0	46.4	23.3	5.8	(25)
(26)		MCWB	3.7	10.5	26.8	39.3	51.1	63.0	67.6	65.1	55.1	38.7	20.5	4.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	22.0	36.8	51.6	61.2	71.4	74.7	72.5	64.7	50.6	31.2	21.1	(27)
(28)		MCDWB	20.1	26.0	46.1	68.7	78.9	83.9	85.1	80.5	74.4	60.6	33.1	25.1		(28)
(29)		2%	WB	11.3	17.9	32.4	46.3	57.9	68.7	72.4	70.3	61.4	46.1	27.9	16.7	(29)
(30)		MCDWB	13.9	21.5	39.9	59.9	72.5	81.5	82.3	78.8	69.0	54.8	30.4	19.9		(30)
(31)		5%	WB	7.4	14.2	29.7	42.8	55.2	66.4	70.7	68.4	59.1	42.6	24.3	8.7	(31)
(32)		MCDWB	9.5	17.1	35.9	54.3	67.0	78.4	80.3	76.7	64.7	50.4	27.6	10.5		(32)
(33)		10%	WB	3.8	10.6	27.0	39.9	52.7	64.4	69.2	66.5	56.7	39.3	20.7	4.2	(33)
(34)		MCDWB	5.4	13.2	32.5	49.9	64.1	75.9	77.9	74.0	62.8	45.7	23.3	5.8		(34)
(35)	Mean Daily Temperature Range		MDBR	16.2	20.8	20.9	20.1	21.8	21.9	18.8	19.6	20.5	17.5	15.8	14.1	(35)
(36)		5% DB	MCDBR	17.2	23.5	21.7	29.5	31.8	29.0	26.4	25.6	25.8	23.4	16.3	14.2	(36)
(37)		MCWBR	15.2	20.2	14.8	17.0	15.1	12.1	10.9	12.6	14.3	14.3	13.4	12.4	(37)	
(38)		5% WB	MCDBR	17.1	23.5	20.6	27.2	26.9	24.7	20.6	21.1	21.0	21.1	15.6	14.5	(38)
(39)		MCWBR	15.2	20.2	14.2	16.0	13.5	11.6	10.1	11.3	12.3	13.6	13.1	12.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.249	0.282	0.337	0.492	0.535	0.496	0.477	0.418	0.381	0.355	0.286	0.248	(40)	
(41)		taud	2.155	2.112	2.014	1.779	1.810	2.021	2.127	2.287	2.305	2.191	2.133	2.098	(41)	
(42)		Ebn at Noon	227	255	261	230	229	242	244	254	248	222	203	198	(42)	
(43)		Edh at Noon	20	30	42	62	63	51	46	37	32	28	21	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	393	769	1334	1675	1665	1925	1657	1453	1098	706	428	293	(44)	
(45)		RadStd	15	21	45	149	111	163	178	112	87	50	25	12	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page